# U.S. University Excellence 2026

Relational Data Analysis of Performance & Economics

*Tools Used: Python (Pandas), SQL (SQLite), Seaborn/Matplotlib*

# Project Overview & Objectives

- **Goal:** Move beyond a simple list to understand *why* universities rank where they do.
- **The Approach:** Build a relational database by joining University rankings with State economic metadata.
- **Primary Questions:** Does state wealth influence rank? Which regions offer the best employment?

# Data Pipeline & Methodology

The project followed a strictly documented data engineering workflow:

• Data Extraction: Primary data was sourced from Kaggle; secondary economic data was manually researched and integrated.

• Data Cleaning (Python): * Standardized headers to snake_case.

• Trimmed whitespace from categorical strings (State, Institution Type).

• Enforced numeric types for scores and dates.

• Verified data integrity (checked for Nulls and 0.0 values).

• Storage & Analysis: The cleaned data was migrated into a Relational SQLite Database.

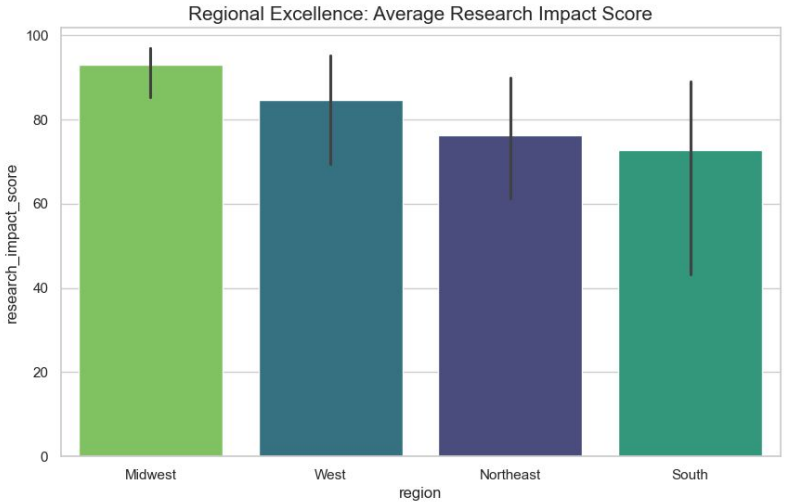
This allowed for complex analysis using JOINs, CTEs, and Subqueries to answer business-critical questions.

# Question 1:

Which U.S. Region has the highest average Research Impact Score?

```sql
SELECT s.region, ROUND(AVG(u.research_impact_score), 2) as avg_research
FROM universities u
JOIN state_info s ON u.state = s.state_code
GROUP BY s.region
ORDER BY avg_research DESC;
```

| | region | avg_research |
|---|---|---|
| 1 | Midwest | 93.0 |
| 2 | West | 84.64 |
| 3 | Northeast | 76.29 |
| 4 | South | 72.83 |



# Question 2:

List all universities located in states where the Median Household Income is higher than the overall national average of all states in our list.

```sql
SELECT university_name, state
FROM universities
WHERE state IN (
    SELECT state_code
    FROM state_info
    WHERE median_household_income > (SELECT AVG(median_household_income)
    FROM state_info)
);
```

| | university_name | state |
|---|---|---|
| 1 | Massachusetts Institute of … | MA |
| 2 | Princeton University | NJ |
| 3 | Stanford University | CA |
| 4 | University of California, Berkeley | CA |
| 5 | Harvard University | MA |
| 6 | Williams College | MA |
| 7 | Yale University | CT |
| 8 | University of California, Los … | CA |
| 9 | Amherst College | MA |
| 10 | University of California, San Diego | CA |
| 11 | California Institute of Technology … | CA |
| 12 | Claremont McKenna College | CA |
| 13 | University of Southern California … | CA |
| 14 | Wellesley College | MA |
| 15 | University of California, Irvine | CA |
| 16 | University of Virginia | VA |
| 17 | Pomona College | CA |
| 18 | University of California, Santa … | CA |
| 19 | University of California, Davis | CA |
| 20 | Boston University | MA |
| 21 | Washington and Lee University | VA |
| 22 | University of Washington, Seattle | WA |

# Question 3:

Find universities in the 'Northeast' that have an Employment Rate higher than 92%.

```
SELECT u.university_name, u.employment_rate, s.region
FROM universities u
JOIN state_info s ON u.state = s.state_code
WHERE s.region = 'Northeast' AND u.employment_rate > 92;
```

| | university_name | employment_rate | region |
|---|---|---|---|
| 1 | Harvard University | 95.0 | Northeast |
| 2 | Massachusetts Institute of … | 96.2 | Northeast |
| 3 | Columbia University | 92.1 | Northeast |
| 4 | Cornell University | 92.6 | Northeast |
| 5 | Princeton University | 94.5 | Northeast |
| 6 | Carnegie Mellon University | 93.7 | Northeast |
| 7 | University of Pennsylvania | 94.8 | Northeast |
| 8 | Yale University | 93.3 | Northeast |

# Question 4:

For each state, show the number of top universities and the average state income, but only for states with more than 2 universities.

```
SELECT s.state_name, COUNT(u.university_name) as uni_count, s.median_household_income
FROM state_info s
JOIN universities u ON s.state_code = u.state
GROUP BY s.state_name
HAVING uni_count > 2;
```

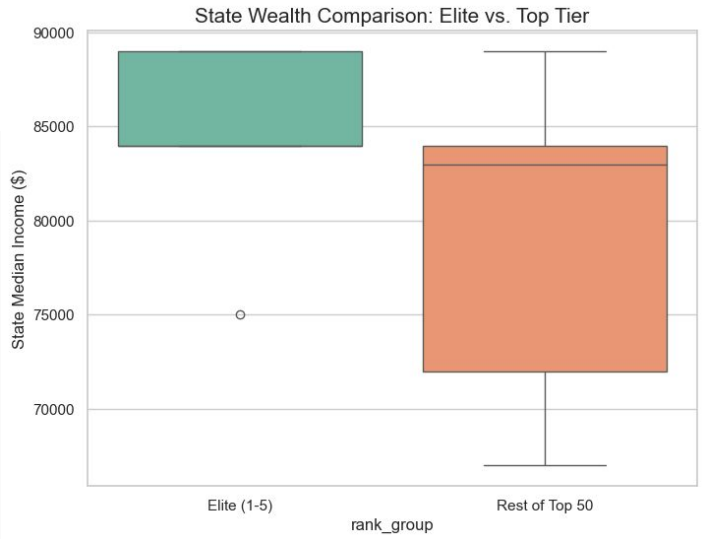
| | state_name | uni_count | median_household_income |
|---|---|---|---|
| 1 | California | 11 | 84000 |
| 2 | Illinois | 3 | 72000 |
| 3 | Massachusetts | 6 | 89000 |
| 4 | New York | 4 | 75000 |
| 5 | Pennsylvania | 3 | 67000 |

# Question 5:

Does being an "Elite" school (Rank 1-5) correlate with being in a high-income state (>$80k)?

```
SELECT
    CASE WHEN u.national_rank <= 5 THEN 'Elite (1-5)' ELSE 'Rest of Top 50' END as rank_group,
    AVG(s.median_household_income) as avg_state_wealth
FROM universities u
JOIN state_info s ON u.state = s.state_code
GROUP BY rank_group;
```

| | rank_group | avg_state_wealth |
|---|---|---|
| 1 | Elite (1-5) | 84200.0 |
| 2 | Rest of Top 50 | 79379.3103448276 |



# Question 6:

Which region hosts the university with the absolute highest International Student Ratio?

```
SELECT u.university_name, s.region, u.intl_student_ratio
FROM universities u
JOIN state_info s ON u.state = s.state_code
ORDER BY u.intl_student_ratio DESC
LIMIT 1;
```

| | university_name | region | intl_student_ratio |
|---|---|---|---|
| 1 | University of Chicago | Midwest | 92.8 |

# Question 7:

Which states in our metadata table currently have zero universities represented in the Top 50 list?

```
SELECT s.state_name, s.region
FROM state_info s
LEFT JOIN universities u ON s.state_code = u.state
WHERE u.university_name IS NULL;
```

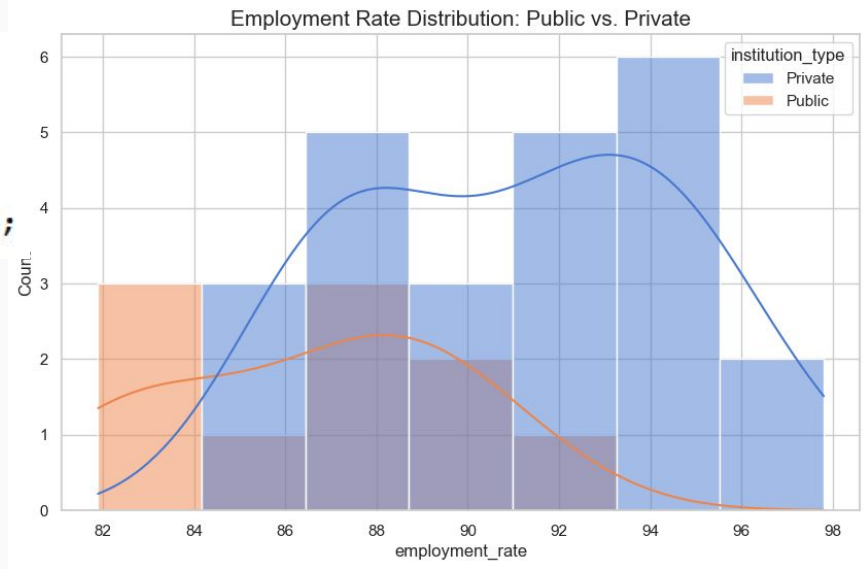
| state_name | region |
|---|---|

# Question 8:

List Public universities that have a higher Employment Rate than the average of all Private universities.

```sql
SELECT university_name, employment_rate
FROM universities
WHERE institution_type = 'Public'
AND employment_rate > (SELECT AVG(employment_rate)
FROM universities WHERE institution_type = 'Private');
```

| | region | active_states_count |
|---|---|---|
| 1 | Midwest | 1 |
| 2 | Northeast | 5 |
| 3 | South | 2 |
| 4 | West | 2 |

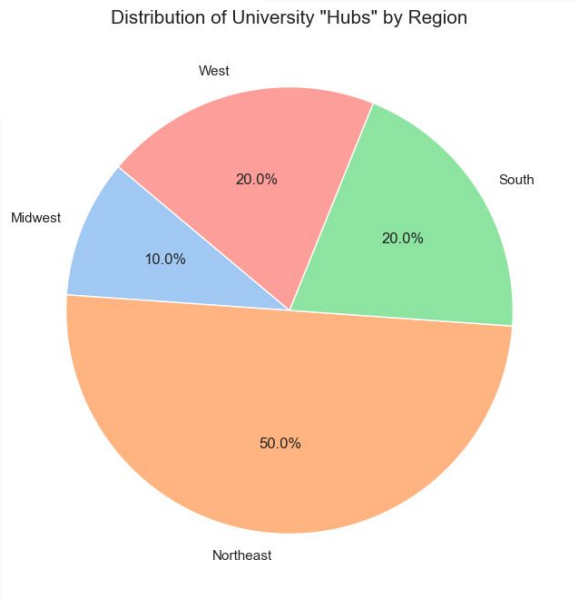


# Question 9:

Whi How many different states in each region have at least one university in the Top 50?

```sql
SELECT s.region, COUNT(DISTINCT u.state) as active_states_count
FROM state_info s
JOIN universities u ON s.state_code = u.state
GROUP BY s.region;
```

| | region | active_states_count |
|---|---|---|
| 1 | Midwest | 1 |
| 2 | Northeast | 5 |
| 3 | South | 2 |
| 4 | West | 2 |



# Question 10:

Create a final report showing each state, its region, the best rank found in that state, and its average research impact.

```sql
WITH StateSummary AS (
    SELECT state, MIN(national_rank) as best_rank,
    AVG(research_impact_score) as avg_impact
    FROM universities
    GROUP BY state
)
SELECT info.state_name, info.region, summ.best_rank,
ROUND(summ.avg_impact, 2) as avg_impact
FROM state_info info
JOIN StateSummary summ ON info.state_code = summ.state;
```

| | state_name | region | best_rank | avg_impact |
|---|---|---|---|---|
| 1 | Massachusetts | Northeast | 1 | 66.12 |
| 2 | New York | Northeast | 2 | 79.08 |
| 3 | New Jersey | Northeast | 3 | 99.0 |
| 4 | California | West | 4 | 83.46 |
| 5 | Texas | South | 12 | 87.75 |
| 6 | Virginia | South | 34 | 57.9 |
| 7 | Washington | West | 50 | 97.6 |
| 8 | Pennsylvania | Northeast | 10 | 78.4 |
| 9 | Illinois | Midwest | 13 | 93.0 |
| 10 | Connecticut | Northeast | 9 | 97.2 |

# Findings & Conclusion

*Economic Correlation: Initial findings suggest that "Elite" institutions (Ranks 1-5) are predominantly located in high-income states (Median Income > $80k), suggesting a feedback loop between state wealth and academic funding.*

*• Regional Dominance: The Northeast continues to lead in Research Impact, driven by a high concentration of Private institutions.*

*• Institutional Balance: While Private schools dominate the top 10, Public universities offer highly competitive employment rates, often outperforming the private school average.*

*The transition from a flat CSV file to a relational database allowed for a multi-dimensional view of the data. By joining institutional performance with state economics, this study concludes that while academic merit is central to ranking, the economic health of the surrounding state provides a critical foundation for a university's global standing.*